

Virtual reality: What is it and should criminologists pay attention?

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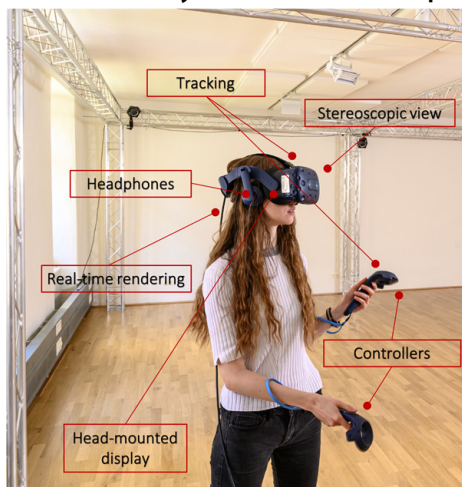
Carefully constructed questionnaires, encompassing observation and interview schemes, large-scale registration data, and cleverly designed longitudinal panel studies have long been commanding approaches to data collection in criminology. The field has made tremendous progress in improving our understanding of criminality through the use of such methods and reached an impressive degree of both qualitative and quantitative sophistication. However, conventional methods are particularly suited to study characteristics of delinquents, such as their dispositions, the families and neighborhoods they come from, their social networks, life-courses, and their educational and criminal trajectories. By design they can offer only limited insight into behavior during crime events and the decision making that underlies those events.

A recent technology that allows for gaining insight into offending processes and, by implication, our understanding of crime and how to prevent it, is virtual reality (VR). In this essay, we provide a brief and non-technical primer on VR and explain what this technology has to offer to criminologists. By way of illustration, we discuss two ongoing criminological research programs using VR. The virtual environments on which both programs are based are made available to the research community to encourage replication efforts and to promote Open Science practices. We conclude by sharing the results of a short survey that was conducted as part of a "VR pop-up booth" set up at the 2023 ASC Annual Meeting and discuss plans to establish a new Division in ASC.

VR: a brief primer

VR refers to interactive and immersive environments that envelope the user and aim to generate the impression that they have stepped inside the virtual world. VR is generally experienced through a head-mounted display (HMD), which provides stereoscopic view allowing for depth perception and three-dimensionality. HMDs are equipped with trackers that trace the location of the user and sends it to the rendering computer. The computer, in turn, feeds the appropriate imagery back to the user through the HMD. This continuous flow of data running from the computer to the user and back, provides the latter with (near) real-time updating, which results in a naturalistic viewing experience (Figure 1).

Figure 1. Virtual reality hardware and basic properties



By shutting the user off from all real-world visual information and replacing it with the computer-generated input, VR perceptually submerges users in the virtual environment, which, if successful, is experienced as effectively real. Although the visual aspect of VR is also its critical feature, it can be complemented with software and hardware that render it more immersive, including spatialized sound which makes users hear sounds as if they are emanating from the surrounding three-dimensional auditory space. Furthermore, devices such as hand-held controllers, gloves, and kinetic suits that provide haptic feedback, and incorporated scents, can add additional layers of immersion to the experience.

The level of immersion a user experiences is often referred to as 'presence' in the VR literature. Presence refers to a user's *subjective* sense of 'being there', i.e., in the virtual environment, and the tendency to respond to virtual events and environments

as if they are real (Slater et al., 2006). The higher the degree of presence experienced by the user, the higher the anticipated correlation between their behavior in a real and an equivalent virtual situation (Bombari et al., 2015). To be sure, the user remains aware that the environment they are seeing is not the real world but, provided certain conditions are met, this does not prevent them from responding to it in realistic ways.

Virtual reality in criminology

Virtual reality technology allows for the creation of realistic situations that closely resemble their real-world analogues and allows for the real-time observation of behavior while providing access to the cognitions, motivations, and emotions underlying it (Van Gelder, 2023). Furthermore, VR allows for developing experimental settings that are easy to manipulate, consistent, replicable, ecologically valid, and rely on hardware and software that are increasingly coming within reach of the median research budget. Whereas these affordances of VR apply to the study of social behavior in general, for research on criminal behavior the technology yields additional potential because crimes generally do not lend themselves to direct observation or manipulation, because there is often an incentive for offenders to misrepresent the facts as they occurred, and because crime is generally too unethical or too risky to study using conventional methods.

Virtual environments can be modeled after contexts that are relevant for crime researchers, such as classrooms, courtrooms, prison settings, bars, streets, neighborhoods, etc. Within such simulated environments, it is possible to stage activities that mimic their real-life equivalents, e.g., a burglary, violent incident, or court proceeding, and implement them as part of assessment, measurement, or intervention strategies (Haberman et al., 2024). By virtue of the possibility to develop criminogenic virtual environments that resemble their real-world counterparts and the possibility of real time measurement of behavior under customizable circumstances, VR allows for the observation of criminal conduct *as it takes place* (Van Gelder et al., 2014; Van Gelder, 2023) rather than having to rely on introspection and retrospection. Furthermore, virtual scenarios conform to the demands placed by controlled lab experiments, yet do not suffer from the problem of low ecological validity or contextual realism that typically characterizes such experiments. By way of illustration of what VR technology has to offer to criminologists, below we briefly discuss two ongoing research programs that make use of virtual reality technology in different ways.

The 360° Virtual Scenario method

The 360° Virtual Scenario Method that is being pioneered by researchers at the Department of Criminology of the Max Planck Institute for the Study of Crime, Security and Law aims to address some of the limitations of traditional written scenarios. This method is based on the assumption that written scenarios are unlikely to capture the more visceral and emotional aspects typically involved in real-world offending and aggression, which commonly occur during “hot” and altered states of mind. Additionally, written scenarios struggle to reflect real-life situations in all their complexity and realistically incorporate important nuances of social experience. Rather than asking people to imagine themselves in a specific situation based on a short description, VR scenarios perceptually immerse them in the situation to enhance the validity of responses.

In recently published work (Van Gelder, 2019, 2022), we developed an immersive 360° video version of a “bar fight” scenario commonly used in criminology (e.g., Exum, 2002; Mazerolle, Piquero, & Capowich, 2003) and compared responses to the virtual scenario with responses to the written version. After either experiencing the conflict in VR or reading about the situation, participants were subsequently asked about their intentions and then responded to a series of survey questions relating to their perceptions and emotions, as well as experienced presence and realism. Overall, results indicate that participants found the virtual scenario to be more realistic, experienced higher presence, and reported more anger, and a higher likelihood of behaving aggressively.

In a larger scale follow up, the MAXLab Aggression and Bystander Intervention Scenario Set (MAXLab_ABISS) was developed to advance the study of decision-making in situations of interpersonal violence (Van Gelder et al., 2024; Herman et al., 2024). MAXLab_ABISS includes scenarios related to aggressive behavior and bystander responses to sexual harassment in barroom settings and allows for addressing a series of research questions related to these phenomena (Figure 2).

Furthermore, it allows for in-the-moment measurement of behavioral intentions, perceptions, and emotions. The linear nature of the scenarios means they can be combined with the real-time measurement of physiological reactions (i.e., cortisol and heart rate) to capture stress responses of participants, as well as eye-tracking. The aim of MAXLab_ABISS is to offer crime researchers a standardized method to study decision making using immersive technology. The stimulus set is freely available for research purposes.¹ For more information, see Van Gelder et al. (2024).

Figure 2. Screenshots taken from sexual harassment scenario of MAXLab_ABISS



Virtual Burglary Project

Another ongoing research program, the Virtual Burglary Project, uses virtual reality versions of residential neighborhoods and houses to study the behavior of burglars. In this research program, incarcerated burglars explore virtual neighborhoods for burglary opportunities or commit a burglary in VR (e.g., Nee et al., 2019; Van Sintemaartensdijk, 2021; McClanahan et al., 2025a). During this process, the VR system tracks their behavior – such as gaze and spatial patterns, which target they select, which entry points they use, and how they go about committing a burglary once inside the target – in real time. In combination with the experimental variation of relevant features of the environment, such as the intensity of street lighting or the presence/absence of guardians, levels of disorder, or house characteristics, this approach provides an unprecedented level of insight into burglary behavior and illustrates some of the potential of VR research to contribute to crime prevention and to inform policy.

A first wave of studies using virtual environments on a laptop computer already showed that burglars tend to perform a burglary in a virtual replica of a house in similar fashion as they burglarize the real house. Furthermore, these studies replicated established findings of earlier research using interviews and surveys among active burglars. In a second wave of studies, immersive virtual neighbourhoods were used to both test theory and the effectiveness of an intervention to deter burglars (e.g., Van Sintemaartensdijk et al., 2021, 2024). These studies focused on the scouting process of burglars and experimentally tested the effect of different levels of guardianship, both physical guardianship with virtual humans acting as guardians, and symbolic guardianship through signs indicating the presence of neighborhood prevention in the neighborhood or increased police surveillance (Van Sintemaartensdijk et al., 2021, 2024). This work, for example, showed that the mere presence of a guardian makes a neighbourhood significantly less likely to be burglarized, but that the behavior of the avatar, hence the level of guardianship, is irrelevant in this context.

A recent study (McClanahan et al., 2025a) that was conducted in prisons in Pennsylvania explored how environmental features influence burglar decision-making. The virtual environment used for this study was modeled on a residential neighborhood typical of Pittsburgh. The researchers recruited incarcerated burglars with varying levels of burglary experience who explored houses with different risk and reward factors related to burglary (e.g., a car in the driveway or a 2nd amendment sign on the lawn). The results show that offenders adjust their perceptions based on these environmental cues, and their proficiency in burglary influences how they assess risks and rewards.

Other current work aims at testing what we term 'dynamic guardianship' in which the environment dynamically responds to the presence of individuals. In different experimental trials, streetlight, sound, and 'Watching Eyes' signs are experimentally manipulated to examine their deterrent effect on burglars, with a view to informing future interventions. The Pennsylvania and dynamic guardianship virtual environments are currently being made available to the research community and can be obtained by sending

an email to maxlab@csl.mpg.de.

ASC 2023 Meeting VR pop-up booth

Criminologists who have had the opportunity to experience VR tend to be enthusiastic about the possibilities the technology offers for their field. At the 2023 ASC meeting in Philadelphia, the authors set up a “VR pop-up booth” at which attendees could experience either a scenario from the 360° Virtual Scenario method or engage with the virtual environment used in the Virtual Burglary Project – Pennsylvania (VBPP) described earlier (Figure 3). In total, 58 attendees at different career stages and from a diversity of institutions, filled out a brief survey about their VR experience. Results of the survey among this (highly selective) sample are reported in Table 1. Respondents found the experience highly realistic, felt strongly immersed in it, and believed the technology holds much potential for criminology and should be incorporated.

Table 1. Responses to VR survey at ASC pop-up booth

Item	Range	Mean	SD
To what extent did this experience feel realistic to you? 1 (not at all realistic) - 5 (very realistic).	2-5	4.34	.690
To what extent did you feel fully engaged or immersed in the presented situation? 1 (not at all immersed) - 5 (highly immersed).	2-5	4.24	.757
To what extent do you believe visual scenarios offer an enhancement over hypothetical written scenarios commonly utilized in criminology and criminal justice research? 1 (not at all) - 5 (very much).	3-5	4.60	.674
To what extent do you believe virtual reality holds potential for advancements in criminology and criminal justice? 1 (not at all) - 5 (very much)	3-5	4.69	.537
To what degree do you believe criminologists should incorporate virtual reality technology into their research practices? 1 (not at all) - 5 (very much).	3-5	4.58	.625



In closing: Promoting Open Science and towards a Division of Virtual Reality within ASC

Virtual reality holds promise as a novel research modality in criminology and we would be thrilled to see it being applied in research across the field and making its way into (under)graduate programs. Given the expertise and costs required for developing virtual environments, collaborations between researchers from different departments seem imperative to achieve this goal. We also see this as a tremendous opportunity. An important advantage of VR is that virtual environments can be easily shared and sent across the globe with a click of a mouse button. This allows for perfect replications, as exactly the same environments are tested by different research groups. However, specific virtual environments can often also be used to address other research questions than the ones they were initially designed for (see McClanahan et al., 2025b, Van Gelder et al., 2024). Virtual environments can, and arguably should, therefore have a shelf life that exceeds a single study. Beyond the scientific imperative of sharing research materials as part of good Open Science practices, investing time and effort into developing high quality environments can also lead to a methodologically stronger field of virtual reality-based research.

With these goals in mind, the authors of this essay have begun discussing ways to improve the collaborative network of VR scholars in criminological spaces. A priority for this group is to establish a Division of Virtual Reality within the ASC that will fulfill various important purposes as virtual reality continues to grow. The proposed Division could, for example, serve as a forum for both established criminologists and new students alike to discuss their VR research, strategies, and findings. The Division will also strive to alleviate the financial and knowledge requirements that are necessary for VR research by channeling resources, such as scholarships and workshops, to those interested in utilizing the technology. Similarly, the Division will work to establish a repository

of simulations (i.e., VR scenarios) that have been used in previous research. This repository would be available to other researchers, allowing for the type of replication efforts that were discussed above. Alternatively, these simulations could be utilized in ways that support entirely new projects and ideas.

In closing, criminologists share an interest in the same or similar outcome variables. Although contexts vary strongly between countries and cultures, there is also much that is shared. For example, the scenarios of the *360° Virtual Scenario method* are situated in an Irish Pub, which are commonly found all over the world and tend to have similar features irrespective of location. This enables the use of the scenarios in different countries. Finally, VR devices are increasingly mobile, which allows for on-site data collection among relevant samples. In the virtual burglary project, for example, researchers set up their field labs in prison facilities and recruit inmates as study participants. For criminologists, this idea of bringing the relevant environment to the target group offers many novel and exciting opportunities.

¹ The scenario set is currently available in the German, Dutch and English language and can be obtained by sending an email to: maxlab@csl.mpg.de

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